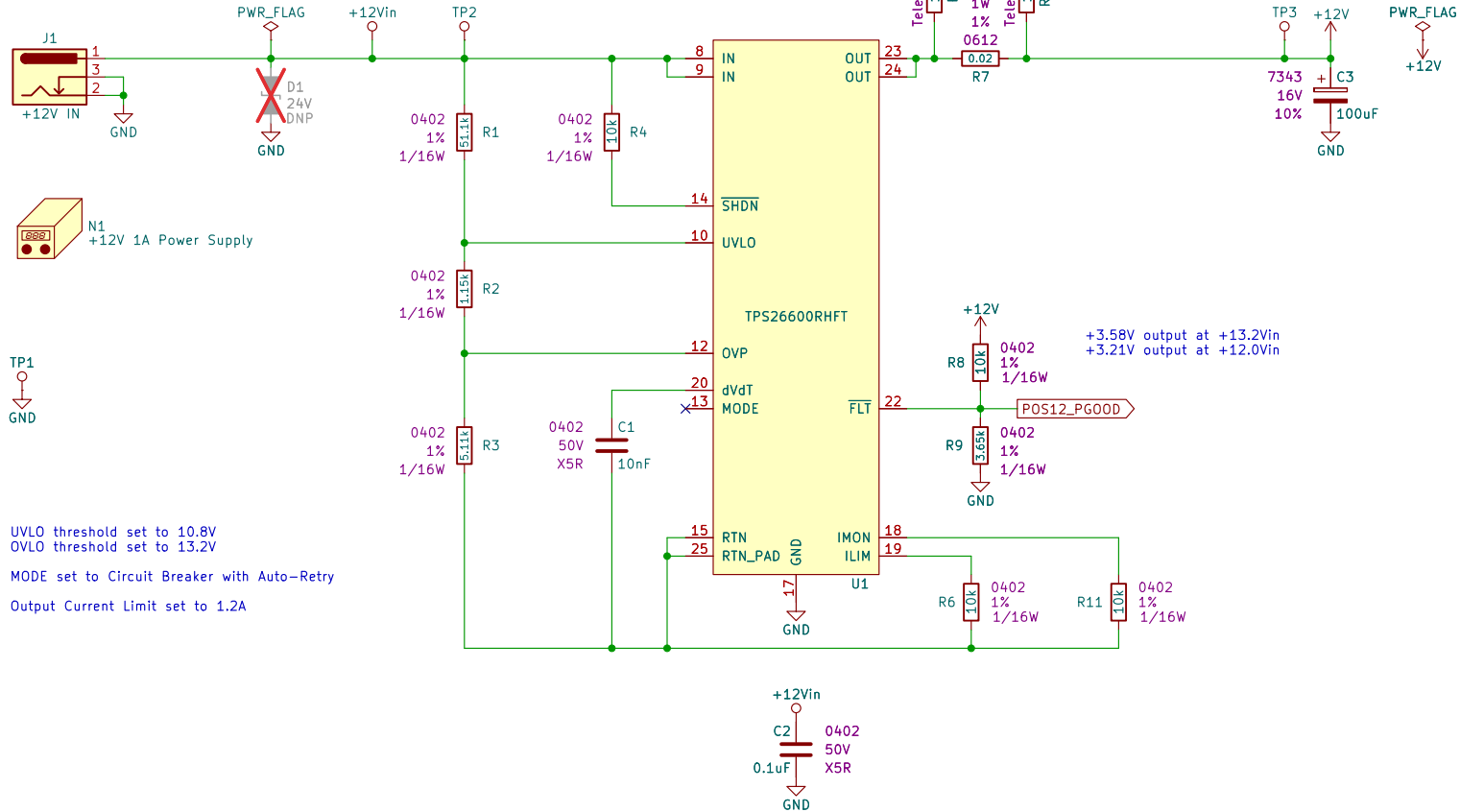


|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 | 5 |
| A | <div>12V Power Input</div> <div></div> <div>File: POS12_power_input.kicad_sch</div> <div>12V Telemetry</div> <div></div> <div>File: POS12_telemetry.kicad_sch</div> <div>3.3V Power Supply</div> <div></div> <div>File: POS3P3_power_supply.kicad_sch</div> <div>3.3V Telemetry</div> <div></div> <div>File: POS3P3_telemetry.kicad_sch</div> <div>dsPIC33 Programming</div> <div></div> <div>File: dspic33_programming.kicad_sch</div> <div>dsPIC33 Power</div> <div></div> <div>File: dspic33_power.kicad_sch</div> <div>dsPIC33 GPIO</div> <div></div> <div>File: dspic33_gpio.kicad_sch</div> <div>USB UART Bridge</div> <div></div> <div>File: usb_uart_bridge.kicad_sch</div> |   | <div>Platform ETC</div> <div></div> <div>File: platform_etc.kicad_sch</div> <div>PGOOD LEDs</div> <div></div> <div>File: pgood_leds.kicad_sch</div> <div>Status LEDs</div> <div></div> <div>File: status_leds.kicad_sch</div> <div>RCA Signal Conditioning</div> <div></div> <div>File: rca_signal_conditioning.kicad_sch</div> <div>Speaker Signal Conditioning</div> <div></div> <div>File: speaker_signal_conditoning.kicad_sch</div> <div>Source Selection, PGAs</div> <div></div> <div>File: source_selection_pgas.kicad_sch</div> <div>Anti-alias Filtering</div> <div></div> <div>File: anti_alias_filtering.kicad_sch</div> <div>Audio Voltage References</div> <div></div> <div>File: audio_voltage_references.kicad_sch</div> <div>LED Panel Connectors</div> <div></div> <div>File: led_panel_connectors.kicad_sch</div> <div>Pushbuttons</div> <div></div> <div>File: pushbuttons.kicad_sch</div> <div>Mechanical</div> <div></div> <div>File: mechanical.kicad_sch</div> |   |   |
| B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |
| C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |
| D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |

|                                                  |                  |          |
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| Sheet: /<br>File: Audio_Visualiser_DSP.kicad_sch |                  |          |
| Title:                                           |                  |          |
| Size: A                                          | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3                              |                  | Id: 1/20 |

# TRY TO TALLY LOAD CURRENTS

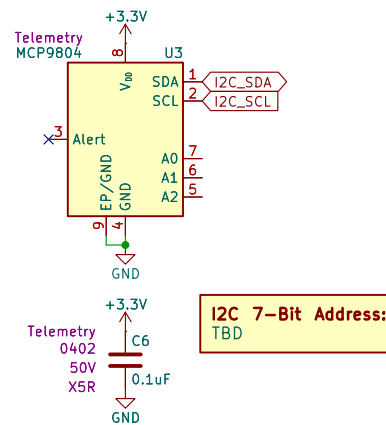
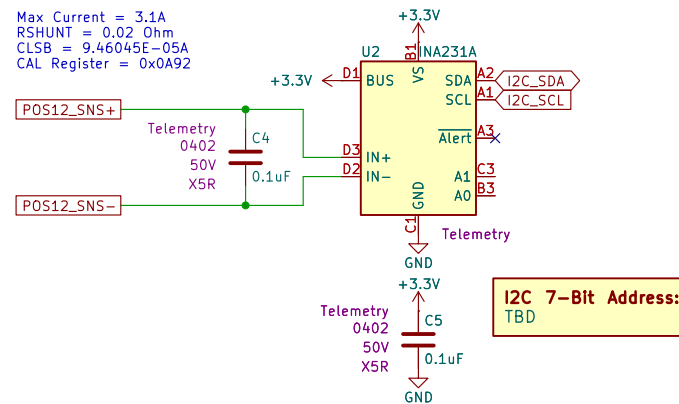


Sheet: /12V Power Input/  
File: POS12\_power\_input.kicad\_sch

## Title:

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

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Id: 2/20



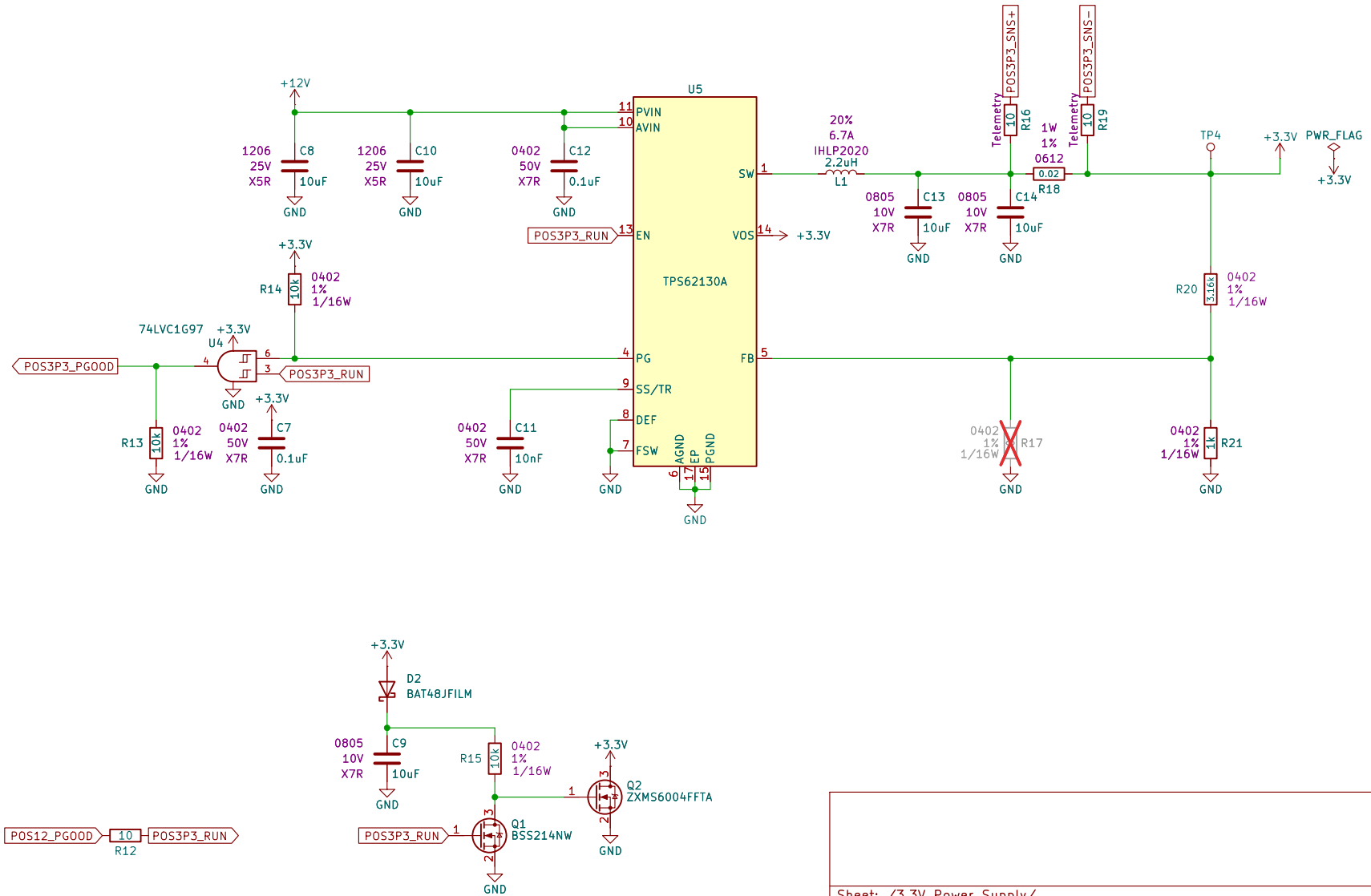
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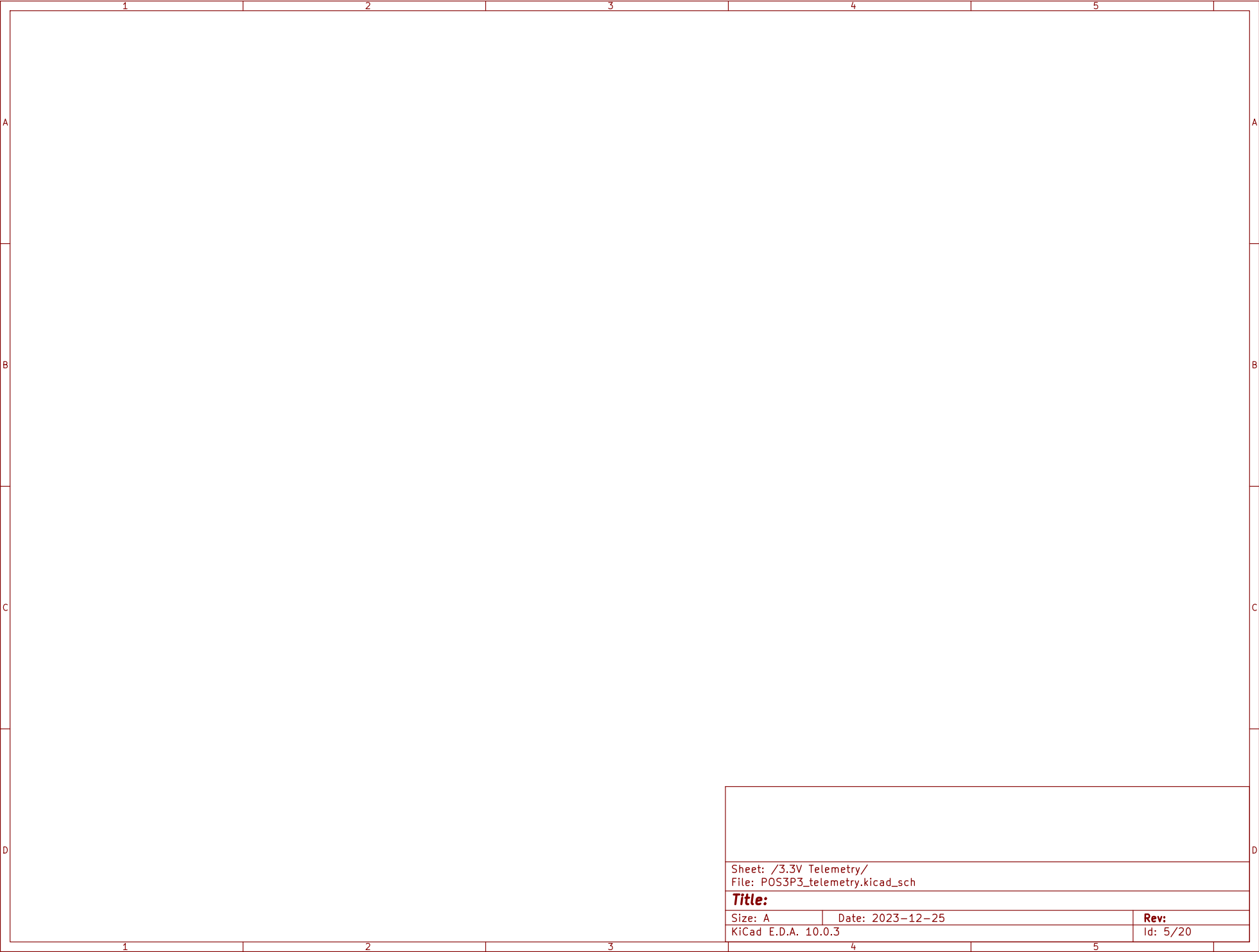
Size: A Date: 2023-12-25  
 KiCad E.D.A. 10.0.3

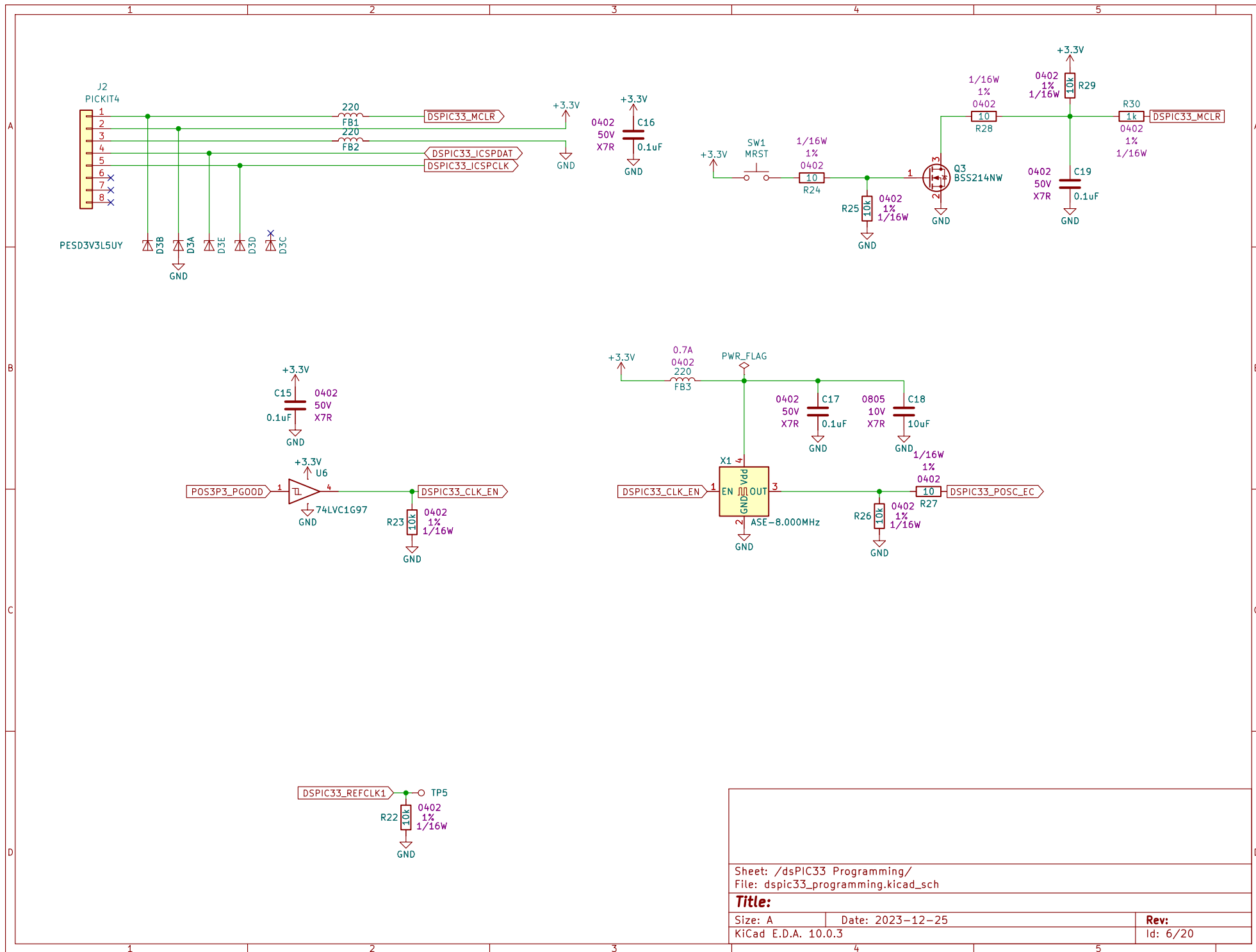
Rev:  
 Id: 3/20

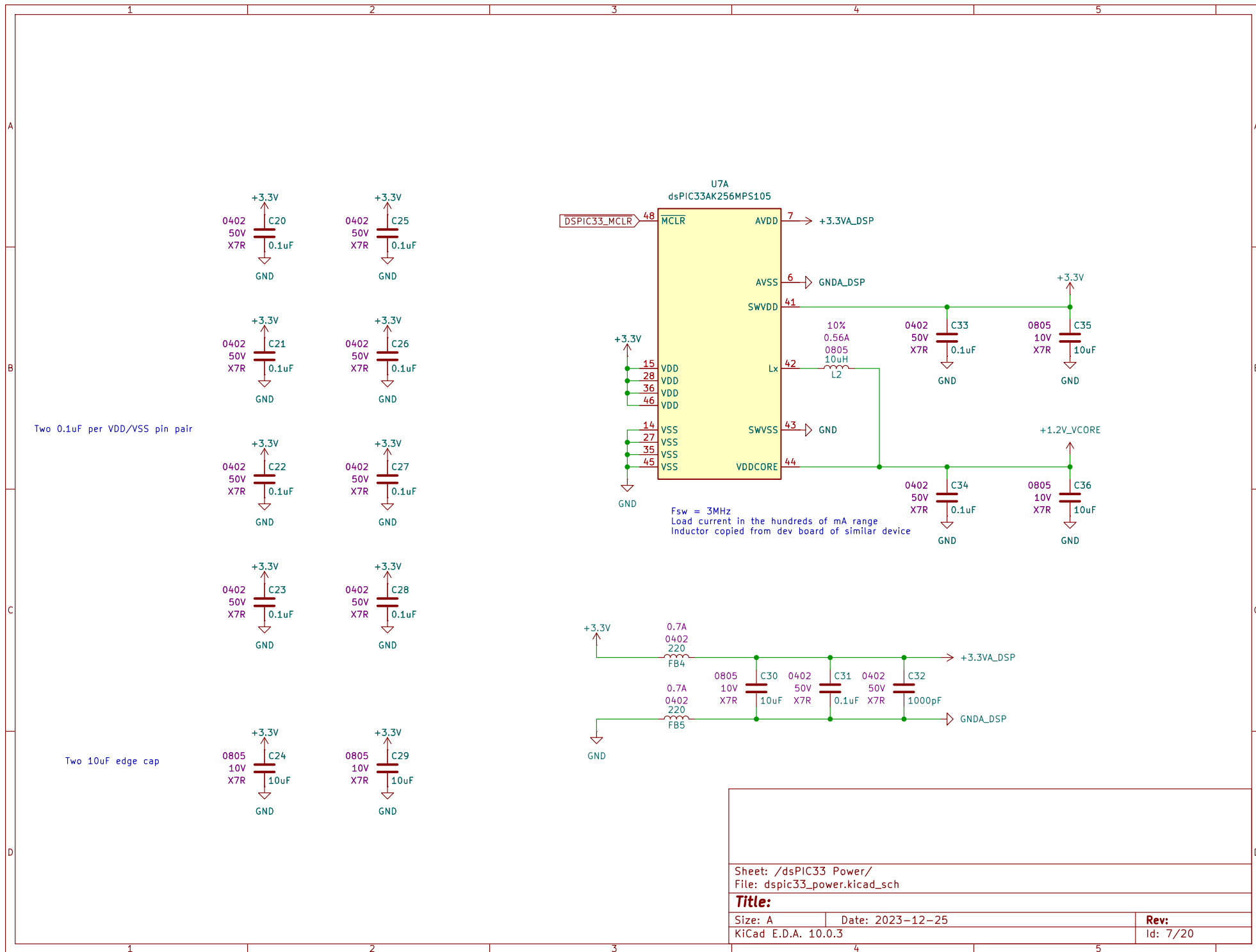
# TRY TO TALLY LOAD CURRENTS



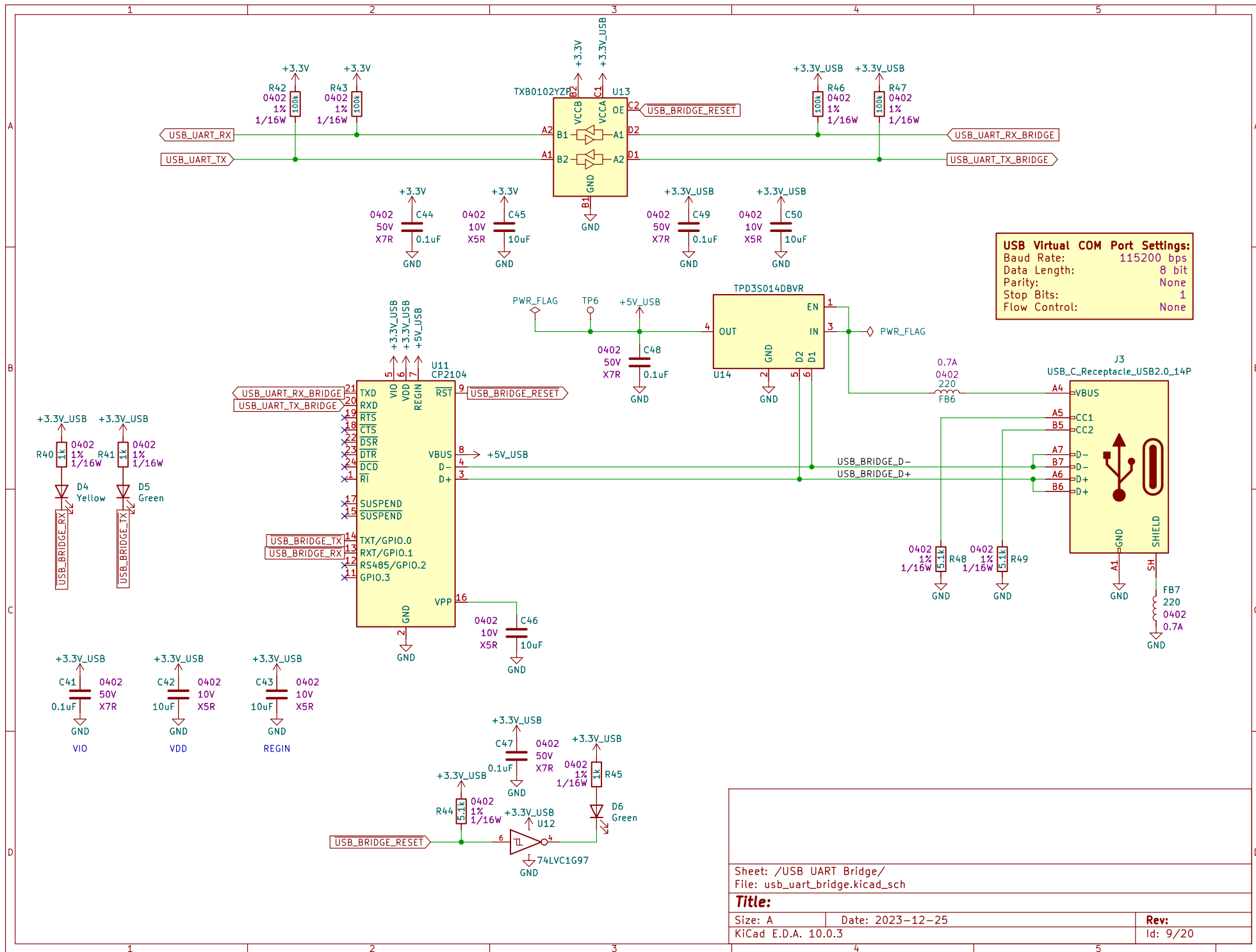
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| Sheet: /3.3V Power Supply/<br>File: POS3P3_power_supply.kicad_sch |                  |          |
| <b>Title:</b>                                                     |                  |          |
| Size: A                                                           | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3                                               |                  | Id: 4/20 |

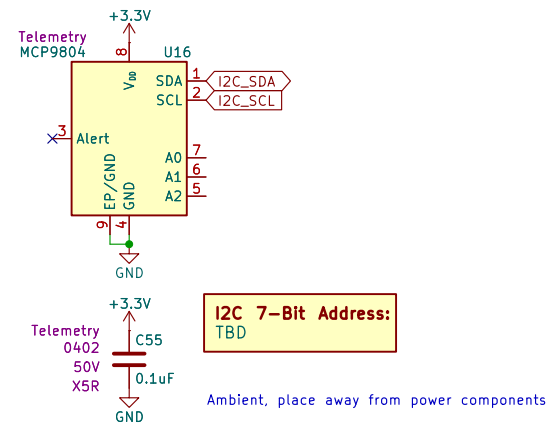
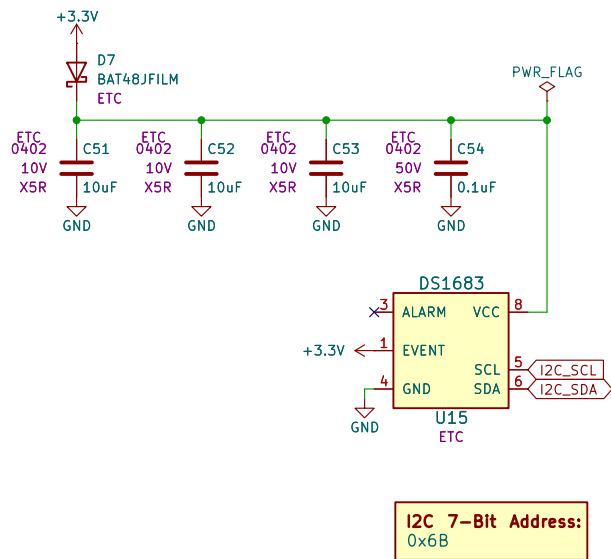










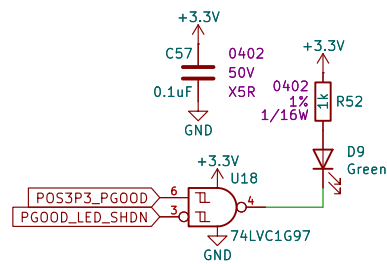
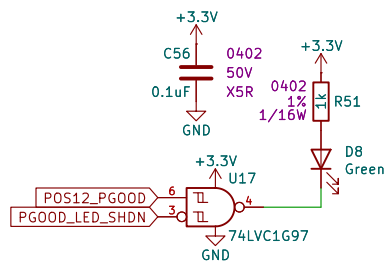
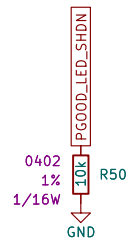


Sheet: /Platform ETC/  
File: platform\_etc.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

**Rev:**  
Id: 10/20

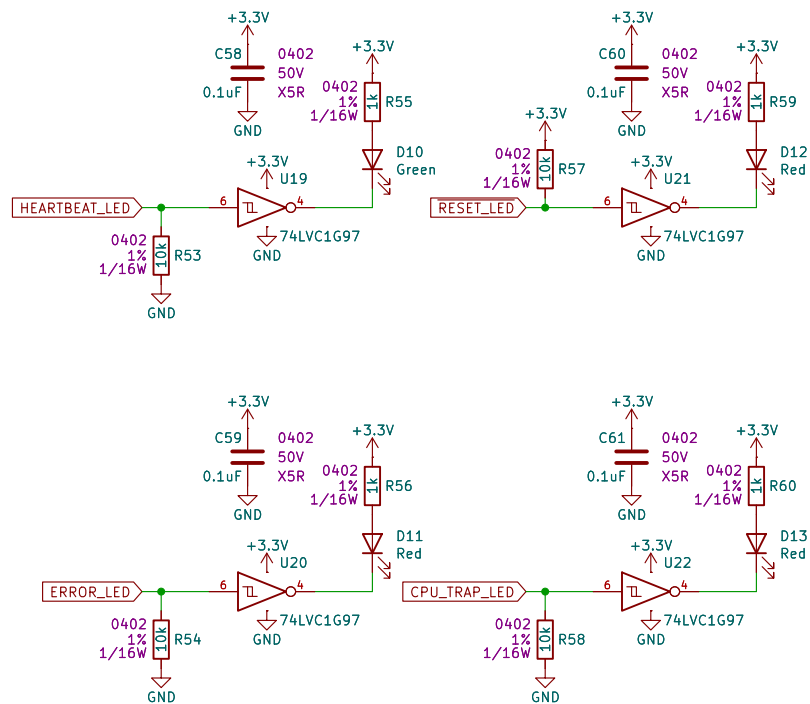


Sheet: /PGOOD LEDs/  
File: pgood\_leds.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

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Sheet: /Status LEDs/  
File: status\_leds.kicad\_sch

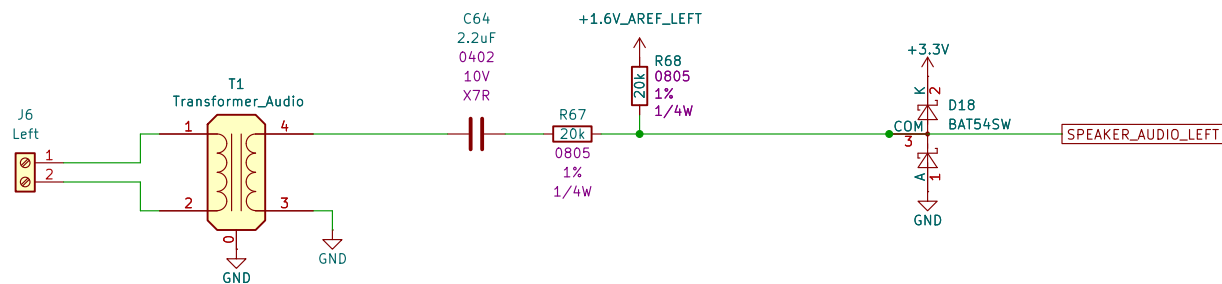
**Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

Rev:  
Id: 12/20



Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp



Sheet: /Speaker Signal Conditioning/  
File: speaker\_signal\_conditioning.kicad\_sch

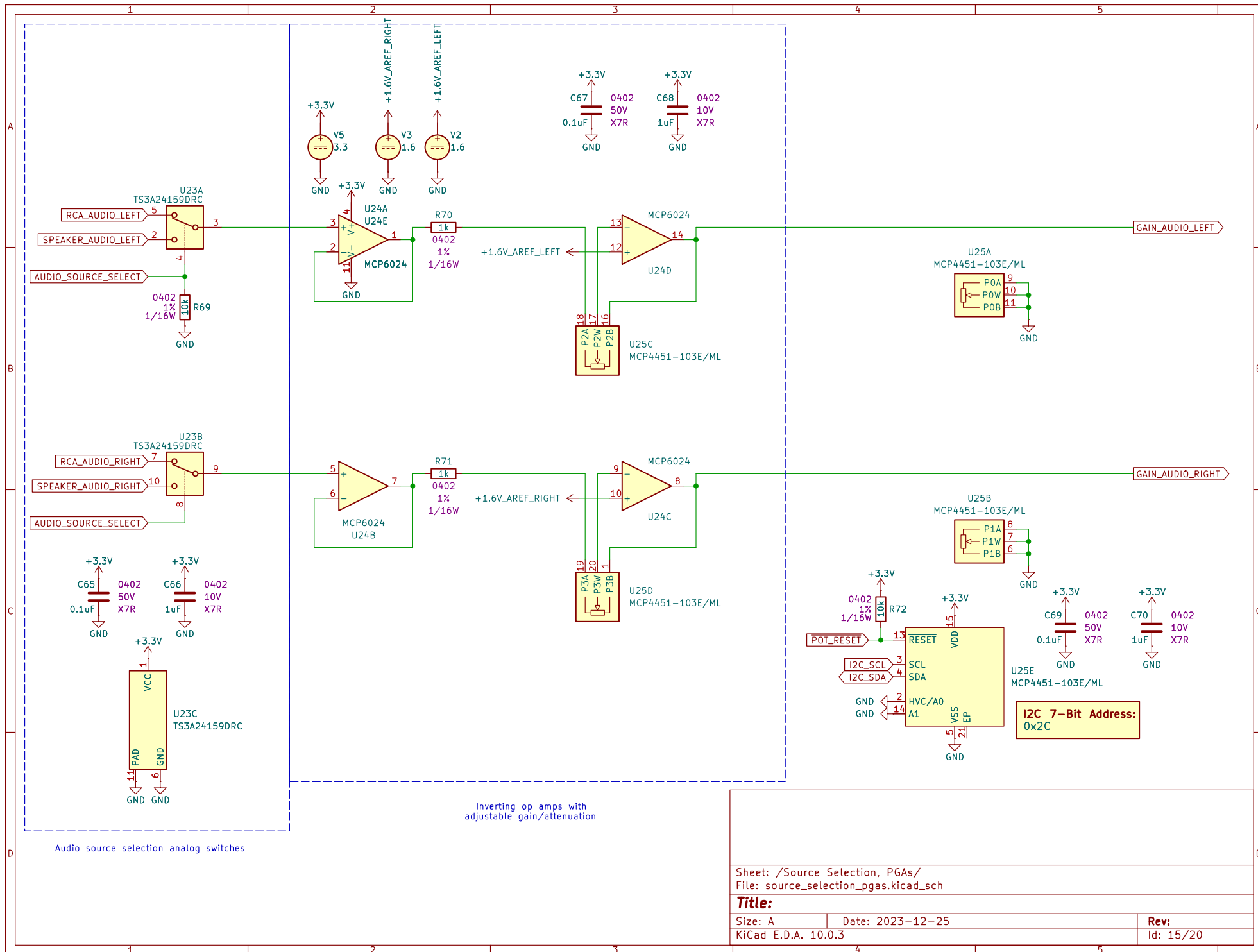
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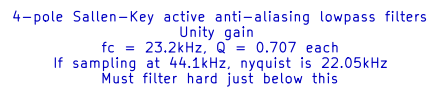
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

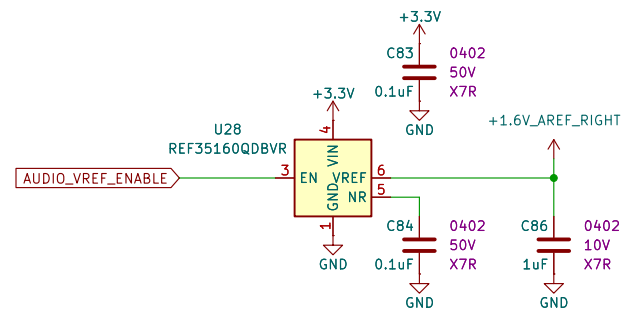
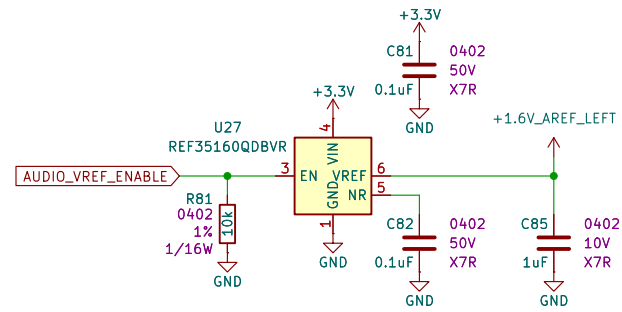
Rev:

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Sheet: /Audio Voltage References/  
File: audio\_voltage\_references.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

Rev:  
Id: 17/20

|   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |
|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

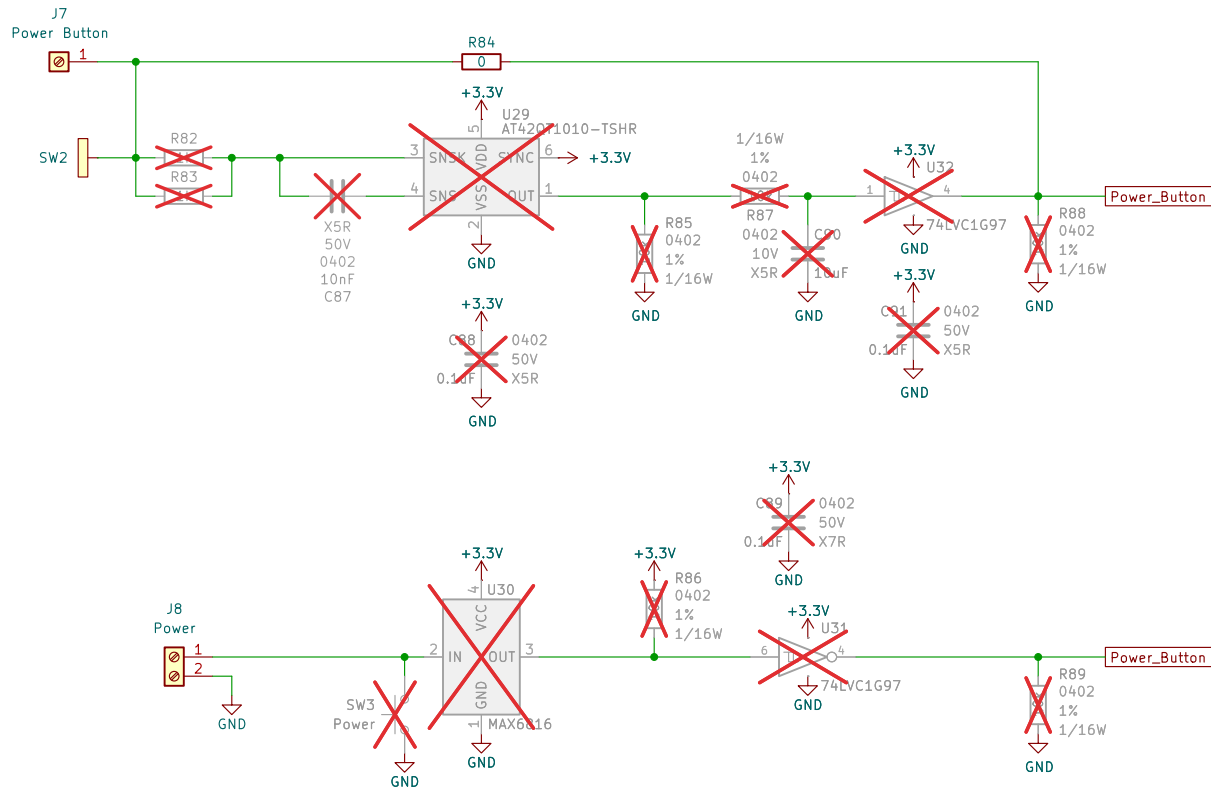
Sheet: /LED Panel Connectors/  
File: led\_panel\_connectors.kicad\_sch

Title:

Size: ADate:KiCad E.D.A. 10.0.3

Rev:Id: 18/20

|                                                                       |       |           |
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| Sheet: /LED Panel Connectors/<br>File: led_panel_connectors.kicad_sch |       |           |
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| Size: A                                                               | Date: | Rev:      |
| KiCad E.D.A. 10.0.3                                                   |       | Id: 18/20 |



Sheet: /Pushbuttons/  
File: pushbuttons.kicad\_sch

**Title:**

Size: A  
KiCad E.D.A. 10.0.3

Date:

**Rev:**

Id: 19/20

